QUANT

Ratio & Partnership, Mixtures & Alligations

Assignment 04

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Ratio& Mixture Online Assignment -04

1. Two casks of 48 and 42 litres are filled with mixtures of wine and water, the proportions in the two casks being respectively 13:7 and 18:17 If the contents of the two casks be mixed, and 20 litres of water added to the whole. What will be the proportion of wine to water respectively in the result?

1. 13:12 2. 12:13 3.21:31 4.31:21

2. In an alloy, zinc and copper are in the ratio 1:2. In the second alloy the same elements are in the ratio 2:3. In what ratio should two alloys be mixed to form a new alloy in ratio 5:8?

1. 3:10 2. 5:6 3.5:7 4.3:7

3. A jar full of whisky contains 40% of alcohol. A part of whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%.. The quantity of whisky replaced is

1. 2/5 2. 1/3 3.2/3 4.3/5

4. How much water must be added to a cask which contains 40 litres of milk at cost price Rs 3.5/litre so that the cost of milk reduces to Rs 2/litre?

1. 24 ltr 2. 30 ltr 3.42 ltr 4.45 ltr

5. How much chicory at Rs 24 a kg should be added to 15 kg of tea at Rs 60 a kg, as to make the mixture worth Rs 39 a kg.?

1. 21 kg 2.20 kg 3.27 kg 4.18 kg

6. A butler stole wine from a butt of sherry, which contained 40 % of spirit. He replaced what he had stolen by wine containing only 16 % spirit. The butt was then of 24 % strength only. How much of the butt did he steal?

1. 2/3 2. 2/5 3.5/9 4. 4/7

7. 729 ml. of a mixture contains milk and water in the ratio 7: 2. How much more water is to be added to get a new mixture containing milk and water in the ratio 7: 3 ?

1. 65 ml 2.81 ml 3.90 ml 4. 100 ml

8. A, B, C enter into partnership, each investing Rs 10,000. After six months, A left the partnership. At the end of the year, a total profit of Rs 30,000 was recorded. Who got the maximum share?

1. A 2. B 3.C 4.B & C both

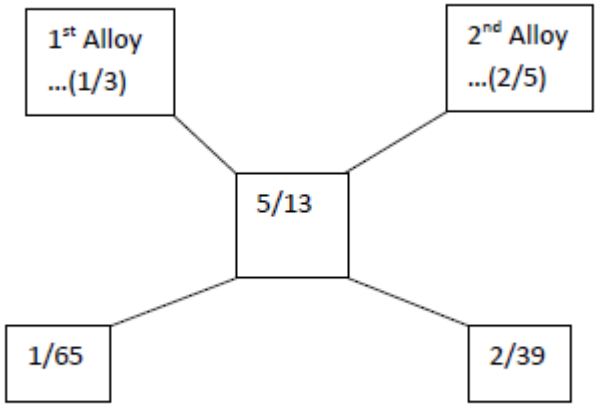
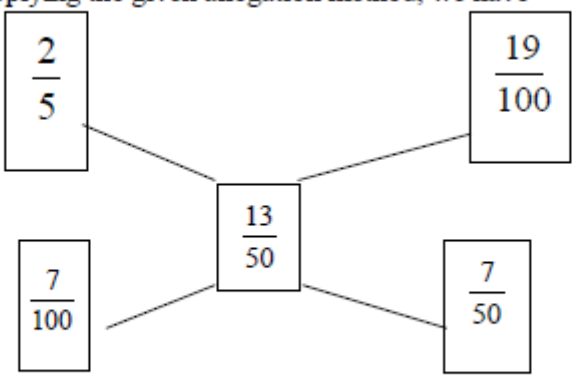
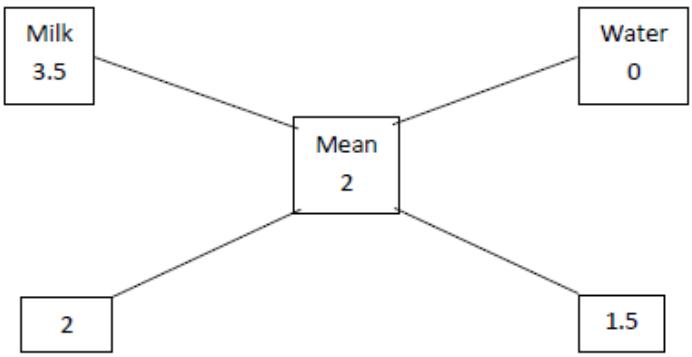
9. X and Y started a business by investing Rs 12,000 and Rs 15,000 respectively. In what ratio the profit earned after 3 years be divided between X and Y respectively?

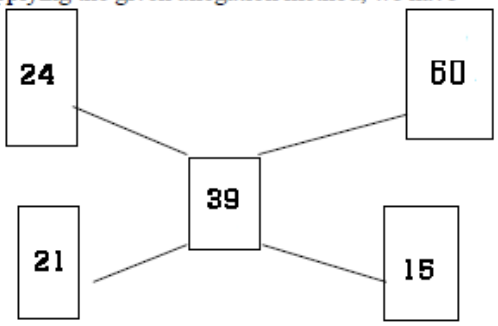
1. 3 : 4 2. 4 : 5 3. 5 : 4 4. 3 : 5

10. Surendra, Rajendra and Mahindra invested some amount in a business in the ratio of 5 : 7 : 6 respectively. In the next year, they increased their investments by 26%, 20% and 15% respectively. The profit earned during the second year should be distributed in what ratio among Surendra, Rajendra and Mahindra?

1. 31 : 27 : 21 2. 21 : 28 : 23 3. 26 : 20 : 15 4. Can't say

Answer key & Explanations

Q No.	Key	Explanations
1.	2	Ratio of wine to water, when 20 litres of water are not added = $(\frac{13\times48}{20}+\frac{18\times42}{35}):(\frac{7\times48}{20}+\frac{17\times42}{35})=$ 264:186 = 44:31. Now, after adding 20 litres of water, Quantity of wine = $\frac{48+42}{44+31}\times44=\frac{264}{5}$ litres and quantity of water = $20+(\frac{48+42}{44+31}\times31)=\frac{186}{5}+20=\frac{286}{5}$ $\therefore$ required ratio = $\frac{264}{5}:\frac{286}{5}=12:13$
2.		Let us consider the fractional values of zinc & copper:  $\therefore$ they should be mixed in the ratio $\frac{1}{65}:\frac{2}{39}$ *or*, $\frac{1}{65}\times\frac{39}{2}=\frac{3}{10}$ *or* $3:10$
3.	3	Now, applying the given allegation method, we have  $\therefore$ ratio of alcohol : whisky = $\frac{7}{100}:\frac{7}{50}=1:2$ $\therefore$ qty. of whisky replaced = $\frac{2}{1+2}=\frac{2}{3}$
4.	2	We can apply the alligation on price of milk, water mixture.  $\therefore$ ratio of milk & water should be 2 : 1.5=4 :3. $\therefore$ added water= $\frac{40}{4}\times3=30$ litres.

5.	1	Now, applying the given allegation method, we have  the ratio happens to be 21 : 15 i.e. 7 : 5. As the tea is given to be 15 Kg.,chicory needed is 7 × 3 = 21 kg.
6.	1	Ratio between wine containing 16% spirit and wine containing 40% spirit, which resulted in wine containing 24% spirit, is 40 – 24 : 24 – 16 $\Rightarrow$16 : 8 $\Rightarrow$2 : 1 ∴ He stole 2/3 of the butt.
7.	2	Quantity of milk in the mixture = $\frac{7}{9} \times 729 = 567$ liters. Hence quantity of water = $\frac{2}{9} \times 729 = 162$ liters. Let x be the quantity of water added. ∴ $\frac{162+x}{729+x} = \frac{3}{10} \Rightarrow x$ = 81 liters.
8.	4	We know, Total investment = Amount invested x number of months Ratio of shares of A, B and C = Ratio of their investments $= 10000 \times 6 : 10000 \times 12 : 10000 \times 12$ $= 1 : 2 : 2$ Clearly, B and C got the maximum share = $\left(30000 \times \frac{2}{5}\right)$ = Rs 12000 Hence, option 4 is the correct answer.
9.	2	We know, Total investment = Amount invested x number of months Both X and Y has invested for 36 months. Ratio of shares of X and Y = Ratio of their investments = 12000 x 36 : 15000 x 36 = 4 : 5 Hence, option B is the correct answer.
10.	2	$5K + 26\%$ of $5K : 7K + 20\%$ of $7K : 6K + 15\%$ of $6K \Rightarrow \frac{630}{100}K : \frac{840}{100}K : \frac{690}{100}K \Rightarrow 63 : 84 : 69$ $\Rightarrow$ 21:28:23